

Competing Risks: Concepts, Methods, and Software

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Annu. Rev. Stat. Appl. 2024. 11:227–54

First published as a Review in Advance on
November 21, 2023

The *Annual Review of Statistics and Its Application* is
online at statistics.annualreviews.org

<https://doi.org/10.1146/annurev-statistics-040522-094556>

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Keywords

subdistribution hazard, cause-specific hazard, cause-specific cumulative incidence, interventions versus biological events, causal versus predictive study questions

Abstract

The role of competing risks in the analysis of time-to-event data is increasingly acknowledged. Software is readily available. However, confusion remains regarding the proper analysis: When and how do I need to take the presence of competing risks into account? Which quantities are relevant for my research question? How can they be estimated and what assumptions do I need to make? The main quantities in a competing risks analysis are the cause-specific cumulative incidence, the cause-specific hazard, and the subdistribution hazard. We describe their nonparametric estimation, give an overview of regression models for each of these quantities, and explain their difference in interpretation. We discuss the proper analysis in relation to the type of study question, and we suggest software in R and Stata. Our focus is on competing risks analysis in medical research, but methods can equally be applied in other fields like social science, engineering, and economics.

1. INTRODUCTION

In the end we all die, but not all at the same age and from the same cause. (Geskus 2020, p. 249)

Competing risks are event-type outcomes that are mutually exclusive. The event can be biological, such as infection, disease onset, or death from a natural cause. It can also be a human intervention, such as organ transplantation, start of treatment, or death from suicide. In both situations, we write that the event “occurs,” although an intervention is typically a human decision.

Sometimes all of the event types are of interest, but more often there is one event type of interest and a competing event may prevent it from occurring. The most classical example is death as a competing risk of some nonfatal event. If the competing risk is a nonfatal event, the event of interest can still occur, but it changes the risk of the event of interest to such an extent that it creates a completely new setting.

Competing risks can be seen as levels of a binomial or multinomial outcome. The probability of each of the competing risks to occur may be related to covariables via a (multinomial) logistic regression. Usually there is a time component involved. This requires the use of time-to-event methods if the time until the occurrence of the event is of interest in itself. Additionally, time may cause the event time to be censored. We restrict this review to competing risks analyses with right censored event times and do not discuss interval censored or doubly censored event times. We briefly touch upon left truncated data due to late entry. Special techniques have been developed for competing risks settings, which extend the ones from standard survival analysis with a single event type. A discussion of these methods is the main topic of this review. We hope that it will help to increase knowledge of these methods such that the presence of competing risks is more often accounted for in the analysis (Austin & Fine 2017).

1.1. Examples

We give a couple of examples that return later in this review, stratified by the role of the competing risks.

1.1.1. Several event types of interest. Event types are of equal interest if we want to investigate whether some factor changes the spectrum of competing event types (see Section 1.1.1.3). A binary outcome represents a competing risks setting if each level represents a different type of outcome. If the time component is not of interest nor relevant, then the two event types are complimentary and thus of equal interest.

1.1.1.1. Treatment for COVID-19. The RECOVERY randomized controlled trial (RCT) has been one of the most influential studies investigating the efficacy of treatment in hospitalized patients with COVID-19. One of the comparisons was between dexamethasone and standard of care (RECOVERY Collab. Group 2021). The primary outcome was all-cause mortality within 28 days after randomization. Time to hospital discharge within 28 days was defined as a separate secondary outcome. Hospital discharge was seen as a beneficial outcome, representing recovery or cure, and can be seen as a complimentary competing risk. Ten percent of the patients remained in the hospital for more than 28 days. This required the use of time-to-event methods rather than a logistic regression, in which those patients were censored at 29 days. One could make “hospitalized for more than 28 days” into another competing risk. However, it would not make sense as an efficacy outcome, because it combines the beneficial event of discharge and the unfavorable event of death. All patients will ultimately die from COVID-19 or be discharged.

1.1.1.2. Outcomes after nosocomial infection. Hospital-acquired infections are a major concern of health care systems. They can be difficult to treat in humans with an impaired immune system,

especially if the pathogen is resistant to medication, leading to increased mortality. The competing events are infection-related death, other causes of death, and discharge. The time component is of interest as well. It may be that infection with a resistant pathogen does not increase mortality but does increase length of stay in the hospital, i.e., time to discharge (De Angelis et al. 2011).

DOC: death from other causes

1.1.1.3. Spectrum in causes of death. Combination antiretroviral therapy (cART) greatly reduced AIDS-related mortality in HIV-infected individuals. Its efficacy has been shown in RCTs. An interesting question is what HIV-infected individuals will die of if they no longer die of AIDS. Is their spectrum in causes of death and the age-dependent mortality of each cause similar to that of individuals without HIV infection? Side effects of cART may increase the risk to die of causes like cardiovascular disease (CVD).

1.1.2. One event type of interest, intervention as competing risk. The disease course is the topic of interest, but it may be interrupted by some human intervention.

1.1.2.1. Disease outcome before intervention. For many diseases, treatments have been developed that greatly improve disease course. When studying the natural disease course, treatment initiation is a competing risk. One example is the start of cART in HIV-infected individuals (Gras et al. 2013). Another intervention that stops the disease course is transplantation. For example, liver transplantation is a lifesaving intervention in patients with primary sclerosing cholangitis (de Vries et al. 2017).

1.1.2.2. Treatment failure. In the search for the best moment to start cART, Zugna et al. (2012) investigated whether individuals who start cART in the first year after HIV infection (early starters) have a different risk of treatment failure than individuals who start later, during chronic infection. Treatment change and treatment interruption were competing risks of treatment failure.

1.1.2.3. Nosocomial infection. Hospital-acquired infection itself can be the outcome of interest, with discharge as competing risk. For example, infection with *Staphylococcus aureus* is frequent among patients with burn wounds, leading to high morbidity and mortality (Norbury et al. 2016).

1.1.3. One event type of interest, biological event as competing risk. The biological event can be part of the disease course, but it can also be the consequence of a human intervention.

1.1.3.1. Efficacy of treatment. The impact of treatment may differ from its direct efficacy. In an RCT on the efficacy of diethylstilbestrol (DES) estrogen therapy on mortality from prostate cancer, death from CVD was a competing risk. A problem is that DES may increase CVD, and the aim is to disentangle the reduction in mortality from prostate cancer and the increase in mortality from CVD. Data from this study have been used in several methodological papers that are discussed in Section 6.3.3.

1.1.3.2. Side effects of treatment. Side effects of treatment can be a study object in itself. A classical example is the study of late complications of radiation therapy in patients with cancer (Caplan et al. 1994, Bentzen et al. 1995). Individuals who received a lower dose of radiation had a lower risk of developing complications; however, they were more likely to have a cancer relapse and die.

1.1.3.3. Relapse and death. Porta-Bleda (2010, p. 32) studied whether the time from diagnosis of bladder cancer to relapse depends on gender. In the data, death from other causes (DOC) was a competing risk.

T : random variable for time until event (of any type)

F : distribution of T ,
 $F(t) = P(T \leq t)$

$\bar{F}$: survival function,
 $\bar{F}(t) = P(T > t) = 1 - F(t)$

T_k : random variable for time until event of type k

F_k : cause-specific cumulative incidence or subdistribution
 $F_k(t) = P(T_k \leq t) = P(T \leq t, E = k)$

E : random variable denoting event type

b_k : subdistribution hazard

1.1.3.4. Death prevents disease outcome. Besides interventions, biological events like death may hamper the study of the disease course. Elderly people can die from many different causes. When studying the natural disease course from HIV infection to AIDS, some individuals die before developing AIDS. Especially among injecting drug users (IDU), mortality from other causes than AIDS has been high (Geskus 2020, section 2.3).

2. MAIN QUANTITIES

We describe and give formal definitions of the main quantities in a time-to-event setting with competing risks. In the standard setting with a single event type, the time-to-event distribution can be described via the density, the hazard, or the cumulative event probability over time. In a setting with competing risks, four different hazards and three different cumulative event probabilities can be defined.

We first combine all event types into a single composite event type and define a random time variable T that takes values according to this time-to-event distribution in the population of interest. Let F describe this distribution: $F(t) = P(T \leq t)$. We follow the medical field and call F the cumulative incidence instead of using the mathematical name, cumulative distribution function. Its complement, $1 - F(t) = P(T > t)$, is often called the survival function, if the event is not death. We use the notation $\bar{F}(t) = 1 - F(t)$.

Let the random variable T_k represent the time until an event of a specific type k occurs. We define $F_k(t) = P(T_k \leq t)$. F_k is commonly called the cause-specific cumulative incidence. A more technical name is subdistribution, which reflects that the cumulative probability does not go up to one because competing events prevent the event of type k from occurring. Let E denote the random variable that represents the event type that is observed, which is one from a collection $\mathcal{K}$ of competing risks. Without loss of generality, we label the competing event types as numbers from 1 to K , hence $\mathcal{K} = \{1, 2, \dots, K\}$. It is easy to see that the sum of the cause-specific cumulative incidences is the overall cumulative incidence, $\sum_{e=1}^K F_e(t) = F(t)$.

T_k can be written in terms of (T, E) as

$$T_k = \begin{cases} T & \text{if } E = k \\ \infty & \text{if } E \neq k, \end{cases}$$

and we have

$$P(T \leq t, E = k) = P(T_k \leq t).$$

In the setting with right censored data, estimation is often based on the hazard (also called rate or incidence). Its definition differs between discrete and continuous distributions. For a discrete distribution we assume that events can only occur at time points $\tau_1, \dots, \tau_L$, with $\tau_1 < \dots < \tau_L$. With respect to the competing risks hazard, there are two different approaches. The subdistribution approach is based on T_k , and the multi-state approach is based on (T, E) .

2.1. Subdistribution Approach

Definitions and properties in the subdistribution approach are completely similar to the standard setting with a single event type, but with T and F replaced by T_k and F_k . There is a hazard function b_k , called the subdistribution hazard, that uniquely describes the distribution. For a discrete distribution, it is defined as

$$b_k(\tau_j) = P(T_k = \tau_j \mid T_k > \tau_{j-1}) = P(T_k = \tau_j \mid T_k \geq \tau_j).$$

Using the law of conditional probabilities, it directly relates to F_k via the product formula

$$\begin{aligned}\bar{F}_k(t) &= 1 - F_k(t) = P(T_k > t) \\ &= \prod_{\tau_j \leq t} P(T_k > \tau_j \mid T_k > \tau_{j-1}) = \prod_{\tau_j \leq t} [1 - b_k(\tau_j)].\end{aligned}\tag{1}$$

λ_k : cause-specific hazard

Equation 1 forms the basis for the product-limit estimator of the cause-specific cumulative incidence in Expression 11. For continuous distributions, the subdistribution hazard is defined as

$$b_k(t) = \lim_{\Delta t \downarrow 0} \frac{P(t \leq T_k < t + \Delta t \mid T_k \geq t)}{\Delta t}.$$

It relates to the cumulative scale via

$$\bar{F}_k(t) = \exp\left[-\int_0^t b_k(s) ds\right].$$

Although it is attractive for its similarity with the standard time-to-event setting, the subdistribution hazard's main disadvantage is that it is not a rate in the classical sense (Andersen & Keiding 2012). Individuals who experience a competing event remain in the risk set, although they can no longer have the event of interest. We can see this by rewriting the subdistribution hazard in terms of (T, E) . For discrete distributions we obtain

$$\begin{aligned}b_k(\tau_j) &= P(T_k = \tau_j \mid T_k \geq \tau_j) = \frac{P(T_k = \tau_j)}{P(T_k \geq \tau_j)} \\ &= \frac{P(T = \tau_j, E = k)}{P[T \geq \tau_j \text{ or } (T < \tau_j, E \neq k)]} \\ &= \frac{P(T = \tau_j, E = k)}{\bar{F}(\tau_{j-1}) + \sum_{e \neq k} F_e(\tau_{j-1})}.\end{aligned}\tag{2}$$

We see what is different from the cause-specific hazard: The term in the denominator contains not only the probability to be event-free until τ_j , but also the probability to have experienced a competing event at an earlier time point.

Sometimes the competing event is not observed. For example, individuals who are cured by cancer therapy are no longer at risk of relapse, but cure itself is not observed. We only observe that some individuals do not develop a relapse. If we keep these individuals in the risk set until the end of their follow-up, then we estimate the subdistribution hazard of relapse in the presence of cure as competing risk. However, often we want to disentangle treatment effects for cured and noncured individuals. Then we would like to remove cured individuals from the risk set, which is what cure models try to do (Amico & Van Keilegom 2018).

2.2. Multi-State Approach

The hazard in which individuals do leave the risk set when they experience a competing event is the cause-specific hazard, which we denote as λ_k . For continuous distributions it is defined as

$$\lambda_k(t) = \lim_{\Delta t \downarrow 0} \frac{P(t \leq T < t + \Delta t, E = k \mid T \geq t)}{\Delta t}.$$

The difference with the subdistribution hazard is in the condition $T \geq t$ instead of $T_k \geq t$: Individuals are no longer in the risk set after an event of any type. It is easy to see that the sum of

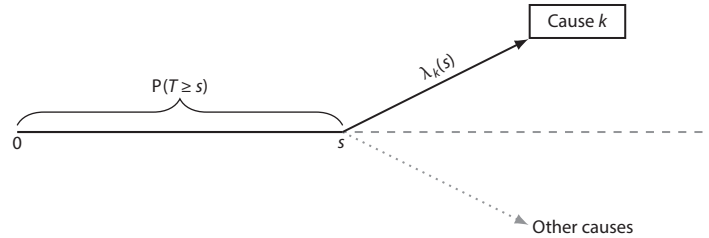


Figure 1

From cause-specific hazard to cumulative incidence. The probability to experience an event of type k by time t is computed based on Equation 3 in the main text. Cause k can occur at any time $s \leq t$. The probability that it occurs at time s is the product of the cause- k specific hazard $\lambda_k(s)$ and the probability $P(T \geq s)$ to remain free of any event until time s . At any time s , other causes can occur as competing risk. Adapted with permission from Geskus (2016).

the cause-specific hazards gives the hazard for all event types combined, the overall hazard (this property does not hold for the subdistribution hazard).

The cause-specific cumulative incidence can be based on the cause-specific hazards via

$$F_k(t) = P(T \leq t, E = k) = \int_0^t P(T \geq s) \lambda_k(s) ds. \quad 3.$$

Figure 1 explains the relation graphically. Equation 3 includes the probability to be free of any type of event $P(T \geq s)$. Hence, contrary to the subdistribution hazard approach, we now need the cause-specific hazards of all competing causes to obtain the cumulative quantity:

$$\bar{F}(t) = P(T > t) = \exp\left[-\int_0^t h(s) ds\right] = \exp\left[-\int_0^t \sum_{e=1}^K \lambda_e(s) ds\right].$$

For discrete distributions the integral in Equation 3 becomes a sum, and it forms the basis for the Aalen–Johansen estimator of the cause-specific cumulative incidence in Expression 7.

The cause-specific hazard is a special case of the transition hazard in multi-state models (Geskus 2016, Putter et al. 2007). Therefore, using the cause-specific hazard as a basis can be seen as the multi-state approach to competing risks.

Since the transition to the cumulative scale depends on all cause-specific hazards, there is no one-to-one relation between cause-specific hazard λ_k and the cause-specific cumulative incidence F_k . Effects on the cumulative scale can even be in the other direction than those on the cause-specific hazard. Geskus (2016, p. 187) explains this by example. It can also be understood intuitively by the phrase “In the end we all die, but not all at the same age and from the same cause” at the beginning of this article and the example of spectrum in causes of death in Section 1.1.1.3. AIDS-related mortality decreased considerably after 1997 due to the introduction of cART. Even though better treatment options decreased cancer-related mortality after 1997 as well (cause-specific hazard), cumulatively we may see more cancer-related deaths if HIV-infected individuals no longer die of AIDS.

2.3. Type of Estimand in the Presence of Competing Risks

Table 1 lists the four hazards and the corresponding cumulative quantities that may be of interest or play a role when analyzing data in the presence of competing risks. The cause-specific hazard, the subdistribution hazard, and the cause-specific cumulative incidence are the main quantities in a competing risks analysis. The overall hazard and overall cumulative incidence define the overall probability to remain event-free in Equation 3. One may also choose to combine competing

Table 1 Hazards and corresponding cumulative quantities in the presence of competing risks

	Hazard	Cumulative quantity
Event types separate	Marginal (λ)	Marginal cumulative incidence [$F^m(t)$]
	Cause-specific (λ_k)	No corresponding quantity
	Subdistribution (b_k)	Cause-specific cumulative incidence [$F_k(t)$]
Event types combined	Overall (b)	Overall cumulative incidence [$F(t)$]

risks into a single composite event and use the overall hazard or cumulative incidence as quantity of interest. The marginal hazard and marginal cumulative incidence describe the time-to-event distribution for a specific event type if, contrary to the real-world situation, the competing event types were absent. With so many quantities to choose from, which one should be the focus in our analysis? We discuss this in more detail in Section 6.

3. DATA STRUCTURE

Before we move to the estimators, we first discuss the time-to-event data structure in a setting with competing risks. In the uncommon situation that we have complete information, all individuals have been followed from the time origin until the occurrence of one of the competing events, and we know the event type. Often, the time component causes information on the outcome to be right censored. Individuals who are still in the study and event-free at the date of analysis are called administratively censored (this is called progressive type I censoring in Andersen et al. 1993). If the study continues, their outcome may be observed at a later time. Another type of right censored data occurs if an individual left the study before the event occurred, which is called loss to follow-up. The event may have occurred already at the date of analysis, but it is not observed.

We need to make an assumption with respect to the unobserved event time in right censored individuals. The main assumption for validity of the standard estimation methods, which are usually based on the hazard, is that censoring is unrelated to progression to the event of interest. This means that right censored individuals at time t have the same residual time-to-event distribution as the ones that remain in event-free follow-up at that time. Individuals who are censored can then be represented by the ones that remain in follow-up in the estimate of the hazard. Unfortunately, the observed data on event and censoring time do not allow us to show correctness of this assumption (Tsiatis 1975; Geskus 2016, section 1.5). Independence can only be argued based on subject-matter knowledge, possibly with the help of additional information.

In the example of outcomes after nosocomial infection (Section 1.1.1.2), loss to follow-up occurs if patients were transferred to another hospital and data from that hospital are not available. If only those who had recovered from the infection were transferred, then we know that they did not die from the infection. If we treat them as censored with respect to other causes of death and discharge, then we assume that they have the same residual mortality and discharge risk as the ones who recovered from the infection but remained in follow-up in the original hospital at that time of transfer. This may be too strong an assumption if transfer is related to severity status of the disease for which they were admitted.

Another form of right censored event times arises when a competing risk prevents the event of interest from occurring. Contrary to the other two types of censoring, the event of interest will never be observed. The interpretation of censoring from a competing risk and how to incorporate it in the estimator differs between a marginal analysis and a competing risks analysis.

When performing a marginal analysis, we hypothesize a setting in which the competing risks are absent. Only in that hypothetical world can we conceptualize a distribution of time to the

event of interest as the single possible event type. The standard estimation methods assume that individuals with an observed competing event can be represented by the ones that remain in follow-up. This assumption is often violated if censoring is due to a competing event. Usually there are unmeasured factors that affect the distribution of both time to the event of interest and time to the competing event. If the competing event is an intervention, the intervention may be made because the individual is progressing to the event of interest (transplant given as a life-saving intervention) or is relatively healthy (transplant given to those with the highest probability of surviving). The assumption is also violated if the exposure or the biological mechanism that leads to the event of interest is, at least partly, responsible for the competing event to occur. Only if all variables that affect both the probability to be censored due to a competing event and the probability of the event of interest have been measured and included in our model can the independence assumption be made conditionally on these variables. However, we never know for certain that we measured and included all these variables. Note that even in an RCT, we need to control for such variables if we want to estimate the marginal distribution.

Sometimes a reasonable alternative approach is to combine the competing risks into a single composite endpoint. Liver transplantation is given as a life-saving intervention in the patients with more severe primary sclerosing cholangitis. When looking for risk factors for mortality, we assume that those who received a transplant were about to die and perform an analysis on the overall distribution that combines death and transplant into a composite endpoint (de Vries et al. 2017). An alternative approach is to build a model that imputes the death time in those who receive transplant (Van Geloven et al. 2017). Not all composite endpoints are a combination of competing risks. Relapse or progression is often combined with death as relapse-free or progression-free survival. They do not prevent death from occurring. The reason to combine them is that they can act as surrogate outcomes that often occur much earlier than death.

If we perform a competing risks analysis, censoring from a competing risk is interpreted as a separate event. In the estimator of the cause-specific hazard (Section 4.1), censored individuals leave the risk set because they experience the competing event. No assumption is made about representation by those that remain in follow-up and no assumption is needed about when they would have had the event of interest. In the estimator of the subdistribution hazard (Section 4.2), the competing event is ignored. Individuals with a competing event remain in the risk set, which reflects that they will never experience the event of interest.

Another type of incomplete information is due to entry into the study after the time origin. It may cause some individuals to be missed because they already experienced the outcome before they could enter the study. Individuals with a shorter event time are more likely to be missed. This length-biased sampling is called left truncation. Estimation in the presence of left truncated data is hardly more involved than with right censored data. We briefly mention it in Section 4, and more details are provided by Geskus (2011) (see also Geskus 2016, chapter 2).

Information on the event type may be incomplete as well. For example, some individuals may have died from an unknown cause. We do not cover imputation of unknown event type in this review. For further details, readers are directed to, e.g., Goetghebeur & Ryan (1995), Bakoyannis et al. (2010), Gouskova et al. (2016), and Lau & Lesko (2018).

3.1. Simulation

The main reasons to simulate data are to understand statistical properties of models and methods, to evaluate and compare performance of estimators, and to perform sample size calculations. Allignol et al. (2011) give an example of how simulation helps to enhance understanding and interpretation of observed results in a competing risks setting.

We give three ways to simulate time-to-event data in the presence of competing risks. At some point in the algorithm, a full probability distribution [i.e., $F(\infty) = 1$] is specified, from which event time data are drawn. Sometimes these data are directly generated from some parametric distribution. In more complicated settings we can often make use of the inverse transform sampling method. This method starts with drawing a random number U on the interval $[0, 1]$, after which an event time is obtained based on the inverse of the cumulative distribution evaluated at u .

Each of the three methods below generates completely observed competing risks data. If the purpose of the simulation is to understand properties of a model or result, we often do not need to consider a setting with incomplete data. If the purpose is to evaluate performance of an estimator or perform a sample size calculation, then we need to consider the real-world setting with right censored and possibly left truncated data. Such data can be added by generating data from a chosen time-to-censoring distribution and time-to-entry distribution. If an individual's event time is larger than his censoring time, he is right censored. If an individual's entry time is larger than his event time, he is excluded from the sample.

3.1.1. Latent failure time approach. The latent failure time approach used to be a common way to describe competing risks data and was often used for simulation (Beyersmann et al. 2009). It defines a joint distribution of all competing event times. Although from a mathematical perspective this is fine, the interpretation in the real world is problematic. The multivariate distribution describes a hypothetical setting in which all event types occur in each individual, which is even more difficult to conceptualize than the marginal distribution. Yet, it can still be used as a basis for simulation of competing risks data. Since the observed data can represent both dependent and independent competing risks, it is easiest to generate event times from independent marginal distributions for each event type and use the smallest one as the observed value.

3.1.2. Multi-state approach. The multi-state approach starts with making a choice for each of the cause-specific hazards λ_k . Since a cause-specific hazard does not define a time-to-event probability distribution, it cannot be used directly to generate data. Instead, we first draw event times t_i from the overall distribution that combines all event types and is based on the sum of the cause-specific hazards $b(t_i) = \sum_i \lambda_i(t_i)$. At this event time, an event is of type k with multinomial probability $\lambda_k(t_i)/b(t_i)$. Details are provided by Beyersmann et al. (2009) and Beyersmann et al. (2012, section 3.2).

3.1.3. Subdistribution approach. In the subdistribution approach, we can restrict to generating time-to-event data for a specific event type k of interest. It starts with choosing a full time-to-event distribution F^* and a probability p that an event of type k occurs [$p = P(E = k)$], which combine into a subdistribution via $F_k(t) = pF^*(t)$. Next, we use inverse transform sampling. If the draw u from the uniform distribution is larger than p , then the event is not of type k . If we are only interested in events of type k , then we can set this competing event time at a very large number—larger than all generated event times of type k . If we want to consider all competing events, then we first specify $p_e = P(E = e)$ for each of the event types and split up the interval $[0, 1]$ over the p_e . This approach was used by Fine & Gray (1999) (see also Beyersmann et al. 2012, section 5.3.6).

3.1.4. Software. Although it is not difficult to write code for generating competing risks data, it has been incorporated in software as well. The R `survsim` package has a function `crisk.sim` (Moriña & Navarro 2017). In Stata the `survsim` command can be used (Crowther 2022).

4. NONPARAMETRIC ESTIMATORS

This section gives nonparametric estimators of each of the three competing risk quantities: the cause-specific hazard, the subdistribution hazard, and the cause-specific cumulative incidence (see **Table 1**). Both hazard estimators can be used as a basis for estimation of the cause-specific cumulative incidence. The estimator of the cause-specific hazard in Expression 6 forms the basis of the classical Aalen–Johansen estimator $\widehat{F}_k^{\text{AJ}}$ in Expression 7. The estimator of the subdistribution hazard in Expression 8 gives rise to a product-limit estimator $\widehat{F}_k^{\text{PL}}$ in Expression 11. With a special choice of the weights that define the size of the risk set r^* , $\widehat{F}_k^{\text{PL}}$ is algebraically equivalent to both $\widehat{F}_k^{\text{AJ}}$ and a weighted empirical cumulative distribution function (ECDF) $\widehat{F}_k^{\text{EC}}$ as defined in Expression 12:

$$\widehat{F}_k^{\text{AJ}}(t) = \widehat{F}_k^{\text{PL}}(t) = \widehat{F}_k^{\text{EC}}(t) \quad \text{for all time points } t. \quad 4.$$

Let N be the sample size. The complete information for individual $i \in \{1, 2, \dots, N\}$ is (t_i, e_i) . The event may be unobserved due to censoring at $c_i < t_i$ (this excludes censoring due to a competing event that is interpreted as a competing event). Define $x_i = \min\{t_i, c_i\}$ and let $\delta_i = I(t_i \leq c_i)$ be the indicator of whether an event was observed. An individual may enter the study event-free after the time origin, at time v_i . The observed data can be represented as

$$\{(v_1, x_1, e_1 \delta_1), \dots, (v_N, x_N, e_N \delta_N)\}. \quad 5.$$

If all individuals are followed from the time origin onwards, then $v_i = 0$ for all and v_i can be ignored. In the following, we only consider this situation without left truncation. Readers are directed to Geskus (2011) for the setting with left truncation.

Let $t_{(1)} < t_{(2)} < \dots < t_{(n)}$ denote the ordered distinct observed event times (of any type), and $t_{(0)} = 0$. Similarly, let $c_{(1)} < \dots < c_{(n_c)}$ denote the ordered distinct observed censoring times. Let m_j be the number of censorings at $c_{(j)}$. For each $t_{(j)}$, $d_k(t_{(j)})$ denotes the number of observed events of type k at $t_{(j)}$ and $d(t_{(j)})$ denotes the total number of observed events at $t_{(j)}$, $d(t_{(j)}) = \sum_{e=1}^K d_e(t_{(j)})$. We make a distinction between individuals who are observed to be at risk and those who are in the subdistribution risk set. In the multi-state approach, individuals leave the risk set when they are no longer observed to be at risk for experiencing the event of interest, either because they experience an event (of any type) or because they are censored. Let $r(t)$ be the number of individuals still observed to be at risk. In the subdistribution approach, individuals who experience a competing event are no longer at risk for the event of type k , but they continue contributing to the risk set. Let $r^*(t)$ denote the size of the subdistribution risk set, to which individuals who had an earlier competing event contribute as well.

Table 2 summarizes all nonparametric estimators of the quantities that may play a role in a setting with competing risks. If administrative censoring and censoring due to loss to follow-up are independent from progression to each of the event types, then we can use the Kaplan–Meier $\widehat{F}^{\text{PL}}$ to estimate overall cumulative incidence. If, additionally, we can assume that the competing risks are independent, then the Kaplan–Meier $\widehat{F}_m^{\text{PL}}$ is valid for estimating the marginal distribution. The competing risk estimators are explained in detail in the next subsections.

4.1. Cause-Specific Hazard, Aalen–Johansen Estimator

The estimator of the cause-specific hazard is completely standard:

$$\widehat{\lambda}_k(t_{(j)}) = \frac{d_k(t_{(j)})}{r(t_{(j)})}. \quad 6.$$

$\widehat{F}_k^{\text{AJ}}$	: Aalen–Johansen estimator
$\widehat{F}_k^{\text{PL}}$	: (weighted) product-limit estimator
$\widehat{F}_k^{\text{EC}}$	: (weighted) ECDF estimator
ECDF:	
	empirical cumulative distribution function
N:	sample size
t_i	: event time
c_i	: censoring time
x_i	: observation min $\{t_i, c_i\}$
δ_i	: $I(t_i \leq c_i)$
v_i	: entry time
e_i	: event type
$t_{(j)}$	: ordered observed event times
d_k	: observed number of events of type k
d	: total number of events
r	: number observed to be at risk
r^*	: number in subdistribution risk set

Table 2 Estimators of hazard and cumulative incidence in the presence of competing risks

Hazard	Estimate	Estimate cumulative incidence
Marginal ^a	$\hat{\lambda}(t) = d_k(t)/r(t)$	$\hat{F}^{m, PL}(t) = \prod_{t_{(j)} \leq t} [1 - \hat{\lambda}(t_{(j)})]$
Cause-specific	$\hat{\lambda}_k(t) = d_k(t)/r(t)$	$\hat{F}_k^{AJ}(t) = \sum_{t_{(j)} \leq t} \hat{F}^{PL}(t_{(j)} -) \hat{\lambda}_k(t_{(j)})$
Subdistribution	$\hat{b}_k(t) = d_k(t)/r^*(t)$	$\hat{F}_k^{PL}(t) = \prod_{t_{(j)} \leq t} [1 - \hat{b}_k(t_{(j)})]$
		$\hat{F}_k^{EC}(t) = \frac{1}{N} \sum_{t_{(j)} \leq t} \frac{d_k(t_{(j)})}{\hat{\Gamma}(t_{(j)} -)}$
Overall	$\hat{b}(t) = d(t)/r(t)$	$\hat{F}^{PL}(t) = \prod_{t_{(j)} \leq t} [1 - \hat{b}(t_{(j)})]$

^aOnly valid if progression to the competing events is mutually independent.

The Aalen–Johansen estimator of the cause-specific cumulative incidence is based on Equation 3 from the multi-state approach:

$$\hat{F}_k^{AJ}(t) = \sum_{t_{(j)} \leq t} \hat{F}^{PL}(t_{(j)} -) \times \frac{d_k(t_{(j)})}{r(t_{(j)})}, \quad 7.$$

in which $\hat{F}^{PL}$ is the Kaplan–Meier estimator of the overall time-to-event distribution and $t_{(j)} -$ is the time point right before $t_{(j)}$. Only the times $\leq t$ at which events of type k are observed contribute to the sum.

In Expression 6, the occurrence of a competing event is treated as a censored observation. However, it is not interpreted as a censored observation. It is only interpreted as a censored observation when we use it to estimate the marginal hazard (assuming that the independence assumption is valid). In that case, Formula 6 estimates both the cause-specific hazard and the marginal hazard, but the cumulative estimates are different. When estimating the marginal distribution, we use the Kaplan–Meier estimator $\hat{F}^{m, PL}$ instead of the Aalen–Johansen estimator $\hat{F}_k^{AJ}$.

4.2. Subdistribution Hazard, Weighted Product-Limit Estimator

The estimator of the subdistribution hazard replaces the number observed to be at risk r in the cause-specific hazard by the number in the subdistribution risk set r^* :

$$\hat{b}_k(t_{(j)}) = \frac{d_k(t_{(j)})}{r^*(t_{(j)})}. \quad 8.$$

Individuals only leave the risk set r^* when they experience the event of interest or when they are censored, not when they experience a competing event. Although the occurrence of the competing event is ignored, we should take into account that individuals can be censored if the competing event had not occurred. This censoring time is typically not known because the event comes first. Only if all censoring is administrative do we know the potential censoring time, which is determined by the date of analysis. If there is loss to follow-up, we need to estimate the probability to be censored over time after the occurrence of the competing event.

For this we define a time-to-censoring random variable C with distribution function Γ . For the estimate of Γ we reverse the role of censoring and event and use the Kaplan–Meier estimator:

$$\hat{\Gamma}(t) = \prod_{c_{(j)} \leq t} \left[1 - \frac{m_j}{r(c_{(j)})} \right]. \quad 9.$$

Note that any observed event is treated as a censored observation, irrespective of the event type. Based on this estimate we can use multiple imputation: generate a censoring time after the competing event occurred, compute the estimator in Expression 8, repeat the procedure multiple times, and average results (Ruan & Gray 2008). A more elegant approach is to use inverse probability weights (IPWs).

In the IPW approach, an individual who experiences a competing event remains in the risk set with a weight that changes over time according to the estimated probability to remain uncensored. The subdistribution hazard only takes values $\neq 0$ at the observed event times $t_{(j)}$ of type k , and we define the individual contribution ω_i to $r^*(t_{(j)}) = \sum_{i=1}^N \omega_i(t_{(j)})$ as

$$\omega_i(t_{(j)}) = \begin{cases} 0 & \text{if } i \text{ is censored or had event of type } k \text{ before } t_{(j)} \\ 1 & \text{if } i \text{ is event-free and under observation at } t_{(j)} \\ \frac{\widehat{\Gamma}(t_{(j)}-)}{\widehat{\Gamma}(t_{(v)}-)} & \text{if } i \text{ had a competing event observed at } t_{(v)} < t_{(j)}. \end{cases} \quad 10.$$

The weight $\widehat{\Gamma}(t_{(j)}-)/\widehat{\Gamma}(t_{(v)}-)$ can be seen as an estimate of the conditional censoring time distribution $P\{C \geq t_{(j)} | C \geq t_{(v)}\}$. A similar procedure is followed in the presence of left truncated data (Geskus 2011) (see also Geskus 2016, section 2.4).

Expression 8 is used as plug-in estimator in the product form in Equation 1 to obtain a Kaplan–Meier product-limit type estimator with a risk set based on r^* :

$$\widehat{F}_k^{\text{PL}}(t) = \prod_{t_{(j)} \leq t} \left[1 - \frac{d_k(t_{(j)})}{r^*(t_{(j)})} \right]. \quad 11.$$

If all individuals with a competing event are given weights based on Expression 9, ignoring the presence of administrative censoring times, the estimator in Expression 11 is algebraically equivalent to the Aalen–Johansen estimator $\widehat{F}_k^{\text{AJ}}$ (Geskus 2011).

In the comparison of mortality between dexamethasone and standard of care in patients with COVID-19 (Section 1.1.1.1), the Kaplan–Meier estimator was used. Those that were discharged were assumed alive until day 28. This can be seen as using the subdistribution approach to estimation because discharge was ignored as competing risk. Note that only 0.1% was censored before day 28 in their data, hence follow-up information was almost complete until day 28.

4.3. Weighted Empirical Cumulative Distribution Function Estimator

With completely observed competing risks data, the cause-specific cumulative incidence at time t is simply the observed fraction of individuals with the event until time t :

$$\widehat{F}_k^{\text{EC}}(t) = \frac{\#\{t_k \leq t\}}{N} = \sum_{t_{(j)} \leq t} \frac{d_k(t_{(j)})}{N}.$$

With right censored data we use a similar IPW approach as in Section 4.2. We then obtain a weighted ECDF form:

$$\widehat{F}_k^{\text{EC}}(t) = \frac{1}{N} \sum_{t_{(j)} \leq t} \frac{d_k(t_{(j)})}{\widehat{\Gamma}(t_{(j)}-)}. \quad 12.$$

Hence, observed events of type k are weighted by the inverse probability to remain free of censoring until that time. Again, there is a straightforward extension to the setting with left truncated data (Geskus 2011). Using Equation 2, it is easy to derive Expression 11 from Expression 12 and vice versa.

Note that the IPW approach in Sections 4.2 and 4.3 gives algebraical equivalence to the Aalen–Johansen estimator, and the weights are not based on a regression model. Therefore, there is no issue of instability as it may occur in marginal structural models (Petersen et al. 2012).

4.4. Standard Errors and Confidence Intervals

Although all three estimators are equivalent (Equation 4), they give rise to different ways to estimate the standard error (SE). We do not know of any systematic comparison of estimators of the SE based on each of the three approaches. Estimation of the SE based on the Aalen–Johansen estimator from Section 4.1 is not straightforward. Several estimators have been proposed, which were compared by Braun & Yuan (2007). For the weighted PL estimator from Section 4.2, which has the same structure as the Kaplan–Meier estimator, an analogous way to estimate the SE can be used. In a simulation study, it performed slightly better than the one that on average performed best in the simulation study by Braun & Yuan (see Geskus 2016, section 2.5). For the construction of confidence intervals, the same applies as in the setting with a single event type: Those computed on the logarithmic and log-log scale perform better than those computed on the untransformed probability scale.

4.5. Group Comparisons

If we want to compare subgroups, we compute group-specific nonparametric estimates. For Equation 4 to hold by subgroup, the estimate $\hat{\widehat{F}}$ that defines the weights should be estimated by subgroup as well. Formal testing can be based on each of the three competing risk quantities. When nonparametrically comparing cause-specific hazards, we perform the standard log-rank test. In the standard setting with a single event type, it tests the null hypothesis of no difference in time-to-event distribution by subgroup. What is different in the competing risks setting is that it only tests the null hypothesis that the cause-specific hazards are equal; it does not test for a difference in cause-specific cumulative incidence because of the absence of a one-to-one relation between both. Gray (1988) suggested a log-rank test that nonparametrically compares the subdistribution hazards, which does reflect the difference on the cumulative scale.

Since the log-rank test is equivalent to the score test in a proportional hazards model, it loses power when the hazard ratio deviates from proportionality. An alternative is to test directly on the cause-specific cumulative incidence. We can compare the estimates at a specific time point using the estimated SEs as in Section 4.4 (Chen et al. 2020). Bajorunaite & Klein (2008) considered alternatives to the log-rank test that cover the whole observed time range. Dormuth et al. (2023) compared type I error and power of several alternatives to the standard log-rank test for the setting with a single event type. It would be interesting to extend this to the competing risks setting.

4.6. Software

The data structure that is needed for estimation is given in Expression 5. The time-to-event information is contained in three columns (or two if there is no left truncation). The difference with the standard setting is in the last column $e_i\delta_i$, which now takes values $0, 1, \dots, K$. Note that instead of numbers, the software may also allow for character values. In that case it should be made clear which value refers to censored observations.

The Aalen–Johansen estimator has been included in most statistical programs, such as the base *survival* package in R. In the subdistribution approach, the occurrence of the competing events is ignored, but we need to extend the follow-up time for individuals who experience a competing event. Imputation of censoring times has been incorporated in the *kmi* R package. Computation

of the time-varying weights ω_i can be done using the `finegray` function in the `survival` package [see the vignette by Therneau et al. (2023)]. One can also use the `crprep` function in the `mstate` package (de Wreede et al. 2011). In Stata, the `stcrprep` command creates the weights (Lambert 2017). The resulting data set with the weights has the same structure as Expression 5, but the last column has only two values, which denote whether the individual has the event of interest observed or is censored (possibly after the competing event occurred). It can be used in any procedure that is based on the subdistribution hazard, as long as the software allows for the inclusion of time-varying case weights. For example, to obtain the product-limit estimator from Section 4.2 in R, the weights are specified in the `weights` argument of the `survfit` function in the `survival` package. It can also be used to fit a regression model on the subdistribution hazard (Section 5.2).

A visualization of the Kaplan–Meier estimate is sometimes accompanied by a table that provides the number at risk over time (Morris et al. 2019). In the visualization of the cause-specific cumulative incidence, there are two options for the number at risk, based on r and on r^* . Which one to choose depends on the purpose of the table. If one wants to give details on the timing of the events and the observed censorings, it is best to use r and provide rows for the number at risk, number of censorings, and the number of events for each event type. If the purpose is to give an impression of the uncertainty in the estimate, it is better to use r^* , which determines the SE. However, confidence intervals are better suited for displaying uncertainty than numbers at risk.

Gray’s log-rank test can be performed using the score test from a weighted Cox model (see also Section 5.2). An alternative is to use the `cmprsk` or `tidycmprsk` R packages or `sts test` in Stata, which have the log-rank test implemented using a different test statistic that performed slightly better in a simulation study (Geskus 2016, section 2.6).

4.7. Example: Bladder Cancer Relapse

For the example of relapse and death (Section 1.1.3.3), **Figure 2** shows the estimates of cause-specific cumulative incidence for relapse and DOC. There is a pronounced difference by gender. Can the lower risk of relapse for males be explained by the higher mortality from other causes, preventing relapse from occurring? In **Figure 3**, the gender difference is also present in the Kaplan–Meier for relapse $\widehat{F}_m^{\text{PL}}$, based on the marginal hazard estimate in which individuals who die from other causes leave the risk set. Moreover, for both genders the cause-specific cumulative incidence and the Kaplan–Meier are almost the same. What can we conclude from this?

Not much. The Kaplan–Meier estimates the marginal distribution for time to relapse, but only under the strong assumption that DOC and relapse are unrelated. Otherwise it has no interpretation. The time-to-event data do not provide the information to determine the relation between DOC and relapse. It could be that every person who died would have relapsed on the next day. This would be the same as adding the cause-specific cumulative incidences of relapse and death. In this most extreme scenario, which cannot be excluded based on the data, the marginal cumulative incidence for the males would be almost similar to the one for the females. The Kaplan–Meier does give information on the cause-specific hazards [because of the one-to-one relation with $d_k(t)/r(t)$]. Hence, we can conclude that females have a higher relapse-specific hazard than males, but that does not tell us anything about what would happen in the absence of DOC.

The estimate of the cause-specific cumulative incidence is never larger than the Kaplan–Meier. Since $r^*(t) \geq r(t)$, it follows immediately from Equation 11. The reason that the Kaplan–Meier and the estimate of the cause-specific incidence are almost the same is because most of the relapses occur during the first 40 months, while the DOC-specific mortality is much lower in that period.

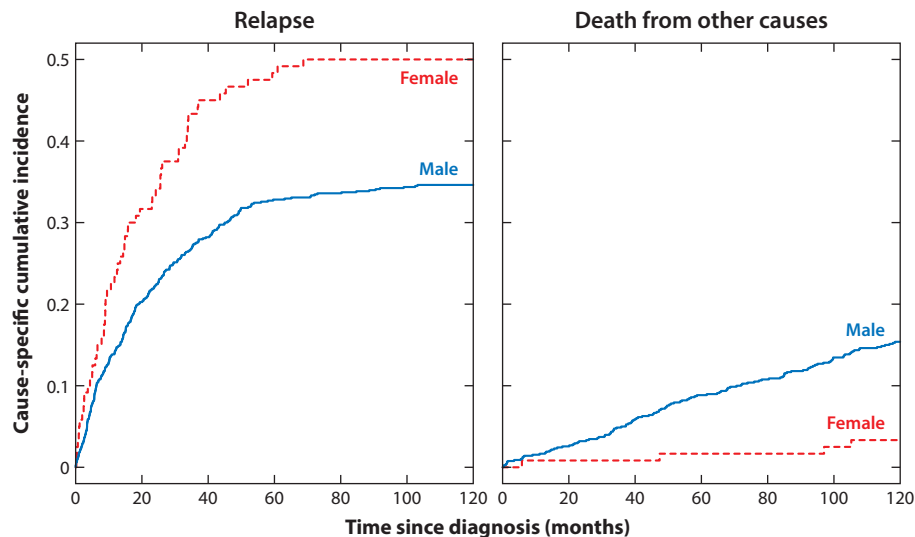


Figure 2

Estimate of cause-specific cumulative incidence for relapse and death from other causes for females and males. There is a pronounced difference by gender. However, the higher mortality among males does not necessarily explain the gender difference for relapse. The data do not provide information on what would have happened with respect to relapse if mortality were absent. Adapted with permission from Geskus (2016).

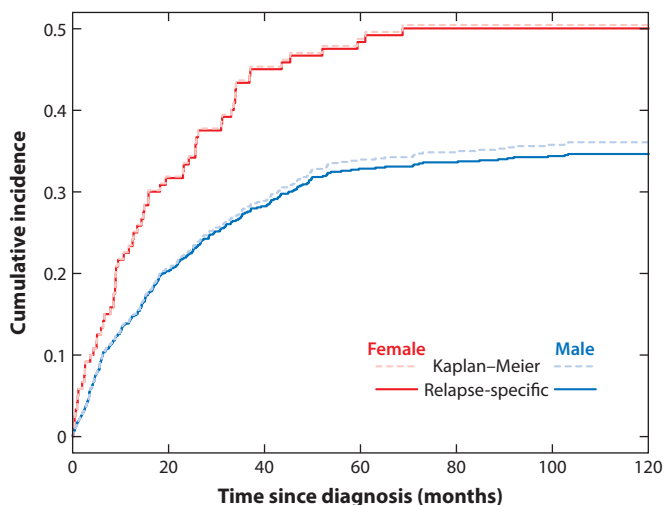


Figure 3

Estimate of relapse-specific cumulative incidence and Kaplan-Meier with relapse as event, for males and females. Both are similar and the gender difference is also present in the Kaplan-Meier. Yet, that does not imply that the gender difference in relapse-specific cumulative incidence cannot be explained by the gender difference in mortality. The Kaplan-Meier may have no interpretation. It estimates the marginal distribution for time to relapse only under the strong assumption that death from other causes and relapse are unrelated. Adapted with permission from Geskus (2016).

5. REGRESSION MODELS

We discuss regression models that specify the relation between covariables and the outcome via either of the three competing risk quantities. If several competing event types are of interest, we can build and analyze a separate regression model for each of them. As explained by, e.g., Andersen et al. (1993), Putter et al. (2007), and Geskus (2016), more information can be retrieved if we combine all event types in a single model and use type-specific covariables. This is not covered in this review. We use $\mathbf{Z}$ to denote the vector of covariables that are included in the regression model.

5.1. Cause-Specific Hazards

Fitting a regression model for the cause-specific hazard boils down to fitting a regression model in the standard setting with a single event type. The reason is that individuals who experience a competing event are treated as censored. Hence, we can use software to fit a standard Cox model if we assume proportional cause-specific hazards. Similarly, if we want to quantify the relation on an additive hazards scale, we can use standard software for Aalen's regression model (Aalen 1989). Lee & Fine (2022) considered an additive hazards model in which the chosen timescale differs by event type (e.g., time from diagnosis for disease-related mortality and age for other causes of death).

Results are to be interpreted in the setting in which the competing risks are present. The estimated relationships do not reflect the marginal hazard, unless we can assume that the competing risks are independent. Just like in the standard setting, including time-varying covariables hardly makes the analysis on the cause-specific hazard more complicated.

We can use results from a regression model on the cause-specific hazard to relate the covariables to the cause-specific cumulative incidence via Expression 3. However, the cause-specific cumulative incidence depends on all cause-specific hazards. Therefore, we first need to fit a regression model for each competing risk. Furthermore, the relation between a variable and the cause-specific cumulative incidence cannot be expressed by a single regression parameter for that variable. The relation depends on the relation between that variable and the cause-specific hazard for each of the competing events, which can even reverse the direction of the relationship on the cumulative scale. Just like in the standard setting, translating relationships to the cumulative scale in the presence of time-varying covariables is considerably more involved.

5.2. Subdistribution Hazard

A regression model for the subdistribution hazard has the advantage that parameter estimates directly translate to the cumulative scale, just like in the standard setting, and we do not need to fit a regression model for each of the competing risks. If only one event type is of interest, analyses can be restricted to that event type. Fine & Gray (1999) introduced a proportional subdistribution hazards model:

$$b_k(t | \mathbf{Z}_i) = b_{k,0}(t) \exp(\beta_k^\top \mathbf{Z}_i).$$

Li et al. (2017) suggested an additive model for the subdistribution hazard.

For estimation in the presence of right censored data, the IPW approach defined in Section 4.2 can be used, which has been extended to the setting with left truncation (Geskus 2011). If the time-to-censoring distribution depends on the same covariables as included in the regression model, we may need to take this into account. For categorical covariables one can compute separate time-to-censoring Kaplan–Meier estimates. For continuous covariables one can use a hazard model for the relation between the covariables and time to censoring, assuming proportionality (Fine & Gray

1999, He et al. 2015) or additivity (Datta & Satten 2002). In a regression setting, it is not clear how to include the probability to remain free from censoring, $\widehat{\Gamma}(t|\mathbf{Z}_i)$, in the weights (Geskus 2016, section 5.3.1). Most authors recommend using the same weights as in Definition 10, but with $\widehat{\Gamma}(\cdot)$ replaced by $\widehat{\Gamma}(\cdot|\mathbf{Z}_i)$ (Fine & Gray 1999, He et al. 2015, Donoghoe & Gebski 2017). A different weight is suggested if Expression 11 is derived from Expression 12.

Fine & Gray (1999) suggested a sandwich-type estimator of the SE that takes account of $\widehat{\Gamma}$ being an estimate of $\bar{\Gamma}$. It was derived by rewriting $\widehat{\Gamma} = \bar{\Gamma} + (\widehat{\Gamma} - \bar{\Gamma})$; the second term gives rise to the sandwich form. However, in a simulation study, it was observed that the regular estimator of the SE performed better in small samples (Geskus 2011).

Inclusion of time-varying covariables is a much more delicate issue than it is for the cause-specific hazard, with respect to both computation and interpretation. Geskus (2020) defined two approaches to estimation, the internal approach and the pseudoindividual approach. In the internal approach, an individual's total follow-up information is seen as coming from that individual. Individuals remain in the risk set after experiencing a competing event. The problem is that we often do not know the value of a time-varying covariable after the competing event, especially if the covariable is an internal one (Kalbfleisch & Prentice 2002) and death is the competing event. Another issue is that the value of the subdistribution hazard at a certain time point depends on the whole history of the time-varying covariables, while the regression model makes the strong assumption that individuals with the same time-instant values of the covariables have the same subdistribution hazard. The pseudoindividual approach circumvents these problems by treating each change in value of a covariable as coming from a different individual, which reduces the data set to one with only time-fixed variables. The pseudoindividual approach assumes that the cause-specific hazards are equal for individuals with the same covariable values. Since the pseudoindividual approach treats all variables as time-fixed, it only quantifies the subdistribution hazard for individuals with constant value of all covariables. This approach was used in a proportional subdistribution hazards model for the change in spectrum in causes of death in HIV-infected individuals after 1997 (Van der Helm et al. 2013).

One approach to prevent the use of time-varying covariables is via a landmark model (van Houwelingen & Putter 2011). For each of a set of follow-up time points, a model is built based on the values of covariables as known at that time point. In a competing risks setting, this may seem a viable alternative for a regression model for the subdistribution hazard with time-varying covariables. However, since the value of the subdistribution hazard at a certain time point depends on the whole history of the time-varying covariables, this approach is problematic as well.

5.3. Cumulative Incidence

The subdistribution hazard is often difficult to interpret, but its main advantage is that parameters for time-fixed covariables have a direct interpretation on the cumulative scale. An alternative is to build a regression model directly on the cumulative scale. Andersen & Ravn (2024) call this a marginal model.

At one specific time point t , either one of the competing events has happened, or an individual is still event-free. We can specify a model on the odds of the occurrence of the event type of interest:

$$\text{logit}[P(T_k \leq t | \mathbf{Z}_i)] = \log[\pi_{k,0}(t)] + \beta_k^\top \mathbf{Z}_i, \quad 13.$$

which translates to the cause-specific cumulative incidence as

$$P(T_k \leq t | \mathbf{Z}_i) = \frac{\pi_{k,0}(t) \exp(\beta_k^\top \mathbf{Z}_i)}{1 + \pi_{k,0}(t) \exp(\beta_k^\top \mathbf{Z}_i)}.$$

IPCW:

inverse probability of
censoring weight

RMTL: restricted
mean time lost

We write $\pi_{k,0}(t)$ as function of t because we can consider the same model at a finite set of time points, but also over the whole time range. This is called a proportional odds cumulative incidence model, because the effect of the covariables is assumed constant over time on the scale of the odds. The effect of time itself is represented by the baseline odds $\pi_{k,0}(t)$.

The proportional odds model and proportional subdistribution hazards model are special cases of a family of semiparametric models that differ in the choice of link function

$$g[P(T_k \leq t \mid \mathbf{Z}_i)] = \alpha_{k,0}(t) + \beta_k^\top \mathbf{Z}_i,$$

where $g(x)$ is the chosen link function and $\alpha_{k,0}(t)$ is an unspecified nondecreasing function (Liu et al. 2018). With $g(x) = \log[-\log(1-x)]$ and $g(x) = \log[x/(1-x)]$, we specify the proportional subdistribution hazards model and proportional odds model, respectively.

For estimation, we need to use weights that correct for the censored data. These weights may depend on covariables. Eriksson et al. (2015) discuss the proportional odds model using estimating equations expressed via the subdistribution hazard that have weights similar to Expression 10. In a simulation study, these estimating equations appeared more efficient than a binomial direct modeling approach that uses estimating equations for the cause-specific cumulative incidence, in which the observed event of interest for individual i is weighted by $1/\widehat{\Gamma}(t_i)$ as in Expression 12 (Scheike et al. 2008). Blanche et al. (2023) compare two different weighting approaches in a logistic regression model for a fixed point t as in Expression 13. One is the binomial direct modeling approach, which they call outcome weighted inverse probability of censoring weight. The other one uses a generalized linear model (GLM) type of estimating equations in which weighting is not performed on the observed events of interest, but on all individuals that are not censored before t via $1/\widehat{\Gamma}(\min(T_i, t))$; they call this approach IPCW-GLM (where IPCW stands for inverse probability of censoring weight). They conclude that the most efficient approach depends on the type of censoring distribution.

A logical extension would be to consider all event types in one model via a multinomial logistic regression that specifies a model on the odds of the occurrence of each of the event types, relative to a reference event type or relative to being event-free:

$$\frac{P(T \leq t, E = k \mid \mathbf{Z}_i)}{1 - \sum_{e=1}^K P(T \leq t, E = e \mid \mathbf{Z}_i)}.$$

Another regression model on the cumulative scale is based on the restricted mean time lost (RMTL) until a chosen time t^* , which is equal to the area under the curve of the cause-specific cumulative incidence. It specifies that $E[t^* - \min(T_k, t^*)] = \int_0^{t^*} F_k(s) ds$ depends on covariables via

$$g\{E[t^* - \min(T_k, t^*) \mid \mathbf{Z}_i]\} = \alpha + \beta_k^\top \mathbf{Z}_i$$

for some link function g . Model coefficients represent the difference in RMTL if g is the identity link and the log ratio if the logarithmic link is used. Estimation can be based on pseudovalues (Andersen 2013) or an IPW approach based on estimating equations with weights $1/\widehat{\Gamma}$ (Conner & Trinquart 2021).

5.4. Flexible Parametric Models

The semiparametric models discussed above leave the baseline function that depends on time unspecified. An alternative is to impose a smooth structure on the baseline function via splines (Hinchliffe & Lambert 2013, Lambert et al. 2017, Mozumder et al. 2021). Such models estimate all components based on a single likelihood function, which has a couple of advantages over the semiparametric approaches. Estimation of SE and confidence intervals is relatively straightforward, mostly using the delta method. And it can easily accommodate time-varying parameters if the proportionality assumption is violated (Kipourou et al. 2019).

5.5. Software

Translating results from a regression model on the cause-specific hazard to the cause-specific cumulative incidence can be done via the R package `mstate` (de Wreede et al. 2011) and the `multistate` module in Stata.

For regression models on the subdistribution hazard, standard software can be used once the data set with weights has been created (see Section 4.6). For example, the Fine & Gray (1999) model can be fitted using software to fit a Cox model, but with specification of the weights. The `crr` function in the R package `tidycmprsk` and the `stcrreg` command in Stata fit the same model, using the sandwich estimator for the SE. In both, creating the data set with weights is incorporated in the procedure.

The proportional odds model can be fitted using the `prop.odds.subdist` function in the `timereg` R package (Scheike & Zhang 2011), or the `logitSurv` function in the `mets` R package. The `comp.risk` function in the `timereg` package uses the binomial direct approach. The two weighting approaches for the logistic regression model, as discussed by Blanche et al. (2023), can be fitted using the `mets` package (functions `binreg` and `logitIPCW`). A regression model on the RMTL using IPW can be fitted via the `rmtl` R package (<https://github.com/s-conner/rmtl>). A regression model on the cause-specific cumulative incidence at specific time points and on the RMTL can also be fitted using pseudovalues, as implemented in the `eventglm` R package (Sachs & Gabriel 2022). Stata allows for the generation of pseudovalues, after which the regression models need to be fitted in a second step (Overgaard et al. 2015).

Flexible parametric regression models have been implemented in the R packages `rstpm2` and `mexhaz`. In Stata one can use the `stpm2cr` module (Mozumder et al. 2017).

6. WHICH TYPE OF ANALYSIS TO CHOOSE?

We turn to what may be the most challenging issues in a competing risks setting: (a) Which quantity do I use in my analysis, and (b) what assumptions do I need to make in order to estimate this quantity? The answers to (a) and (b) relate to the type of study question as well as to the character of the competing risk (see the sidebar titled Points to Consider). Typically the study question involves the relation between covariables and outcome, which is translated into the formulation of a regression model. Li et al. (2022) give an informative flowchart that may help in the decision.

6.1. Three Types of Study Questions

Leek & Peng (2015) distinguish between six types of study questions. In this review article, we focus on inferential, predictive, and causal study questions. Shmueli (2010) makes a similar type of distinction but calls the first one descriptive.

Inferential questions are about association or correlation. The purpose of the analysis is to determine the existence and strength of a relationship, without formalizing the underlying mechanisms. Both Leek & Peng and Shmueli call it the most commonly applied type of analysis.

POINTS TO CONSIDER

1. Is the study question inferential, predictive or causal?
2. Is the competing risk an intervention or a biological type of event?
3. Are we interested in the outcome in the presence of competing risks or marginally?

Predictive questions are about individual outcomes. The aim is to predict the outcome at the individual level as accurately as possible, preferably using information from few and readily available variables. Interventions play a special role in prediction models. They can be included as variables in the model, but they are decided by humans and do not reflect biological mechanisms in a natural setting. If the decision of when to intervene varies by setting and location, the model becomes less generalizable. If all individuals are treated according to the same general guidelines, then current practice can be seen as the natural setting.

Gillies (2019) describes different approaches to causality in medicine. One philosophical school characterizes causality completely in terms of actions, interventions, and manipulations. An RCT quantifies the effect of an intervention on the outcome. Its main purpose is to see whether a new intervention improves disease outcome, not to increase understanding of disease mechanisms, and it often does not make the mechanism explicit. Another philosophical school of causality postulates that causality can only be concluded if we can specify a mechanism that explains the relation. According to the Russo–Williamson thesis (Russo & Williamson 2007), we need both statistical evidence (often through interventions on humans) and evidence of a mechanism (often through laboratory experiments on animals) to infer causality.

6.2. Marginal Analysis

An important distinction is between a marginal analysis and a competing risks analysis (**Table 1**). In a marginal analysis, there is one event type of interest. As explained in Section 3, a marginal analysis is often problematic. Standard approaches to analysis require that the occurrence of the competing risks is unrelated to progression to the event of interest, and this assumption is often violated. We may use background knowledge and correct for dependent censoring via a regression model that includes variables that are known to relate to progression to each of the event types. However, we can never be sure that these variables explain all of the dependence. An alternative is to make assumptions with respect to the relationship, as has been done under the heading of semi-competing risks. The dependence is modeled using frailties or assuming a copula structure (Fine et al. 2001). Although copula models can have a fairly flexible structure (Deresá & Van Keilegom 2023), they are still based on assumptions that cannot be verified based on the observed data. Another problem with considering a marginal distribution is that it may refer to a purely hypothetical setting, especially if the competing risk is a biological event.

If the competing risk is an intervention or a direct consequence thereof, the setting without the competing risk is often imaginable. In order to select patients with primary sclerosing cholangitis that more urgently need transplantation, we need to learn about risk factors for mortality if patients do not receive a transplant (example in Section 1.1.2.1). In order to choose the best time to start cART in HIV-infected individuals, we need to learn about the natural disease course if individuals do not start cART (When to Start Consort. 2009). Both are predictive study questions that refer to the marginal distribution. A causal study question that is answered via the marginal distribution is the comparison of the time from HIV infection to AIDS in IDU and men who have sex with men (MSM) (example in Section 1.1.3.4). Many more IDU than MSM died before AIDS diagnosis. Death before AIDS is a competing risk, and we want to make the comparison in the hypothetical situation that they do not die of other causes. Some causes of death, such as overdose and suicide, can be interpreted as interventions.

In the example of nosocomial infection (Section 1.1.2.3), the marginal distribution is the one of interest if we want to quantify the infection risk in a hospital, possibly in comparison with other hospitals. It considers the setting in which patients are not discharged. This is a causal question, with the hospital as a proxy for the exposure (the real exposure is harder to define but it comprises

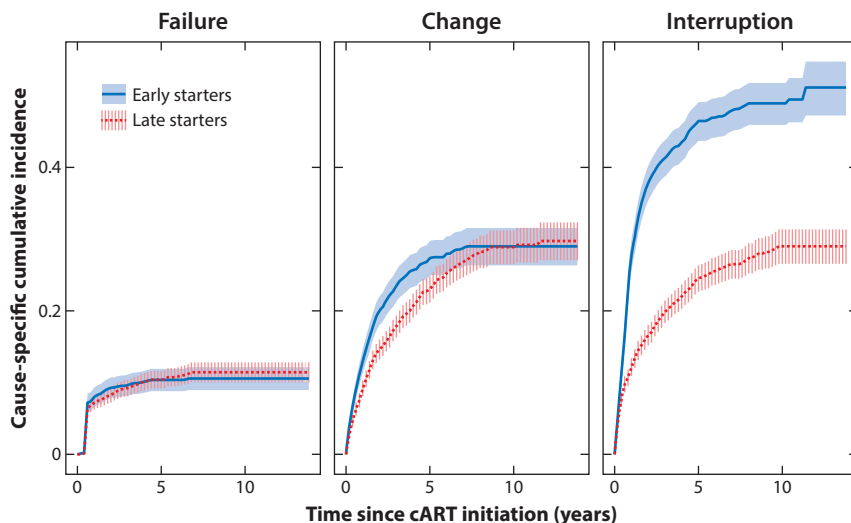


Figure 4

Cause-specific cumulative incidence (with 95% confidence interval) by exposure group. Unexpectedly, the early starters did not have a lower probability of treatment failure. They were much more likely to interrupt treatment, but this does not necessarily explain the similar failure-specific cumulative incidence. The data do not tell us when those that interrupted treatment would have failed had they not interrupted. Adapted with permission from Geskus (2016). Abbreviation: cART, combination antiretroviral therapy.

the conditions that impact infection risk in a hospital). Another question is the financial or disease burden from *Staphylococcus* infection in the hospital. This question of what percentage of patients actually get infected while staying in a hospital is answered in a competing risks analysis. The difference from the marginal analysis above is in the contrast between “while staying in the hospital” (the actual situation) versus “if continuing to stay in the hospital” (a hypothetical situation). Van Geloven et al. (2017) discuss a similar type of contrast between marginal and competing risks analysis for patients on dialysis.

We now discuss the example of treatment failure (Section 1.1.2.2) in more detail. **Figure 4** gives the cause-specific cumulative incidence estimates for the three event types. Contrary to expectation, the early starters did not show a lower probability of treatment failure. However, they were much more likely to interrupt treatment. The marginal study question is what would have happened if nobody had interrupted or changed treatment. Interruption and change are both interventions. However, the independence assumption is questionable. It may be that in both groups, individuals interrupt treatment because they are responding well to therapy. What would have been their failure time without interruption? If the ones who interrupt treatment are less likely to fail, the most extreme scenario occurs when they will never fail. Then the marginal distribution would be the same as the reported subdistribution, and in that case we cannot conclude that there is a difference in risk of failure between the two groups. In order to investigate the influence of potential confounders, a proportional subdistribution hazards model was fitted that included risk group and age. Results did not change much.

In the rare case that the competing risk is a biological event that is unrelated to the event of interest, the setting without that event is imaginable and independence may hold. An example is death from lung cancer or CVD in the comparison between IDU and MSM above. However, the marginal distribution still cannot be estimated in this data because other causes of death, such as overdose and suicide, are likely to be related to progression to AIDS.

Independence may also hold if the competing event is caused by an external, unrelated factor. In the example of nosocomial infection (Section 1.1.2.3), it holds if infection is a random process that does not depend on a patient's severity of disease and thus does not determine the rate of discharge without infection.

In the example of the gender difference in relapse of bladder cancer (Section 4.7), we are inclined to relate observed results to the marginal setting. However, this only makes sense if the factors causing DOC are unrelated to relapse, or if we can include all variables that affect both DOC and relapse in our analysis (age is one of them).

6.3. Competing Risks Analysis

If the competing risk is a biological event, a setting without the competing risk is often irrelevant or difficult to imagine. It is irrelevant for the inferential study question about the impact of the introduction of cART on the spectrum in causes of death in HIV-infected individuals (Section 1.1.1.3). All competing events are of interest and only a competing risks analysis makes sense. It is also irrelevant for a predictive study question if the competing risks cannot be prevented. A setting without competing risks is difficult to imagine if the question is causal and the competing event is a possible outcome of the same exposure or biological mechanism that leads to the event of interest. An example is the efficacy of DES in preventing mortality in patients with prostate cancer (Section 1.1.3.1).

Often the discussion focuses on the choice between a model for the cause-specific hazard and the subdistribution hazard. Latouche et al. (2013) suggest fitting models and reporting results based on both hazards, because they quantify different aspects. The cause-specific hazards model quantifies the relation with the instantaneous event rate, while the subdistribution hazard directly relates to the cumulative scale.

Even though the regression model may be based on the hazard, the cumulative quantity is often more relevant. This is clear when the question is predictive. An individual wants to know the probability of dying within a certain time span, not the probability of dying on a specific day conditionally on being alive. It is also the more interpretable quantity when the study question is causal (Section 6.3.3).

We now discuss the appropriate analysis for each of the three types of study questions if the presence of competing risks is assumed.

6.3.1. Inferential question. Since the aim is to find relationships, it is appropriate to consider different approaches. We follow Latouche et al. (2013) and suggest to fit a model for the cause-specific hazard as well as a model that quantifies associations on the cumulative scale. The latter can be either a model for the subdistribution hazard or a model directly on the cause-specific cumulative incidence. Often it is also insightful to consider relationships with each of the competing risks.

6.3.2. Predictive study question. For prediction we can use any regression model that can translate estimates to an event probability. This can be a model based on cause-specific hazard, on subdistribution hazard, or directly on the cumulative probability, as described in Section 5. Ideally, the one that has the best predictive performance is selected. But ease of use may be a criterion as well. We may favor a model based on the cause-specific hazard because software is readily available to fit such models and to translate results to the cumulative probability. Note, however, that we need to fit a model for each of the cause-specific hazards, with the risk of incorrect model specification, even if only one event type is of interest. Also, the relationship between predictors and outcome is not directly expressed via parameters. This is not an issue for prediction in clinical

practice, but having a direct parametric representation may help in interpretation of results. This is provided by a model for the subdistribution hazard or a model directly based on the cause-specific cumulative incidence.

With respect to evaluating model performance, the setting is similar to the classical setting. Van Geloven et al. (2022) give an overview of performance measures and modern validation methods. Gerds & Kattan (2021) give a general overview of risk prediction models in a medical setting, including the setting with competing risks.

6.3.3. Causal study question. When the study question is causal, there are several issues to consider. We assume the simplest setting, that we have data from an RCT and we want to quantify the effect of an intervention. The correction for baseline and time-varying confounding as present in observational data is the same as in the standard setting with a single event type and is not discussed here.

In the standard time-to-event setting, the predominantly chosen effect measure has been the hazard ratio, estimated via a Cox proportional hazards model. However, any contrast between hazards is problematic, even in an RCT setting (Hernán 2010). The hazard ratio cannot be given a causal interpretation, because the hazard is an event probability, conditionally on being event-free. This condition gives rise to imbalance in risk groups as soon as there is an unmeasured variable that affects the outcome (Stensrud et al. 2019, Martinussen 2022). Measures based on the cumulative incidence or functions thereof, such as the RMTL (Section 5.3), are more appropriate and can be interpreted as causal effect estimates. Note that the regression models can be based on the hazard, but results from these models need to be translated to contrasts on the cumulative scale. In a competing risks analysis, the cause-specific hazard ratio has often been recommended as the closest approximation to an effect measure and to answer etiological questions (Lau et al. 2009, Latouche et al. 2013, Austin et al. 2016, Geskus 2016). However, both the cause-specific hazard and the subdistribution hazard suffer from the same problem with interpretability as the hazard in the standard setting.

Contrary to the standard survival setting, an effect measure based on the cause-specific cumulative incidence in a competing risks setting is problematic as well. The value of the cause-specific cumulative incidence is also determined by the effect of the intervention on the competing event. In the example of side effects of treatment (Section 1.1.3.2), the lower risk of complications in cancer patients who received a lower dose of radiation therapy may, at least partially, be explained by the higher mortality due to cancer. In the example on efficacy of treatment (Section 1.1.3.1), the lower risk of death from prostate cancer in patients receiving DES may also partially be explained by the higher risk of dying from CVD. It is certainly recommended to quantify the relation between the intervention and the probability of both the event of interest and the competing event, but that is not the final answer. Some recent papers have given further insights and have provided possible solutions, which we briefly summarize.

Young et al. (2020) give a detailed description of the causal framework in a competing risks setting and define the possible causal estimands using counterfactuals. They develop a causal directed acyclic graph (DAG) in which the competing event can be seen as an intermediate variable. The contrast with respect to the cause-specific cumulative incidence is a total effect, because it combines the effects through all causal pathways between the intervention and the event of interest. The contrast with respect to the marginal cumulative incidence can be seen as a controlled direct effect, because it quantifies the effect on the event of interest for the setting in which the mediating competing event is eliminated, i.e., set as absent. Our interest is in the natural direct effect, the effect of the intervention that does not incorporate its effect on the competing risk as intermediate variable. Only if there is no effect of the intervention on the competing event,

then the total effect of the intervention on the cause-specific cumulative incidence has a causal interpretation as natural direct effect.

Stensrud et al. (2020) formulate another framework under which the causal pathways can be disentangled. The main assumption is that the total effect on the event of interest can be decomposed into two different mechanisms, one directly affecting the event of interest Y and the other affecting the competing event D . They are called separable direct and indirect effects. This is formalized by extending the causal DAG to an augmented DAG that explicitly splits the effect of intervention A into an effect A_Y and an effect A_D . They give as an example the effect of DES in patients with prostate cancer. The effect of DES is separable if it reduces mortality due to prostate cancer by suppressing testosterone production and increases cardiovascular mortality through estrogen-induced synthesis of coagulation factors. Note that the separation criterion is also an untestable assumption, which must be justified by subject matter knowledge. However, it may give a framework for future interventions in which the components can be set to different values, which can then be tested in a 2×2 factorial RCT.

Another important assumption in this framework is that there are no unmeasured variables that affect both event types. Stensrud et al. (2020) considered the situation with only baseline variables, while Stensrud et al. (2021) generalized this to the setting of time-varying variables that affect both event types. An additional issue with time-varying variables is that their value may be affected by the treatment. Both papers consider a discrete time setting, while Martinussen & Stensrud (2021) consider estimation of separable direct and indirect effects in a continuous time setting.

6.4. Software

We refer to Section 5.5 for translation of results from a regression model on the cause-specific hazard to the cumulative probability. One of the R packages to evaluate predictive performance is `riskRegression`.

Software to implement the methods described in Section 6.3.3 is still in development. Young et al. (2020), Stensrud et al. (2021), and Martinussen & Stensrud (2021) all provide R code as supplemental material. A more user-friendly version of the R code can be found at <https://www.elenadudukina.com/post/competing-events/2022-11-24-competing-events>. Syriopoulou et al. (2022) give a detailed explanation of Stata code using a flexible parametric model and the `g-formula`/regression standardization. All use the data from the study on the efficacy of DES in patients with prostate cancer (data can be found at <https://hbiostat.org/data>).

DISCLOSURE STATEMENT

The author is not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

ACKNOWLEDGMENTS

The author is supported by the Wellcome Trust (grant number 225167/Z/22/Z).

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